CHAPTER III

METHODOLOGY

This chapter presents the Requirements Analysis, System Framework, Technical Background, Implementation Plan, and the Conceptual Framework used for the development of EVENTECH: An Integrated Event Management System and Service Outsourcing Platform for D'Saints Grill and Restobar in San Carlos City, Pangasinan.

3.1 Requirements Analysis

The research design for this study is based on the use and application of both Descriptive and Developmental research designs in addressing the operational gaps identified in the processes and operations of D'Saints Grill and Restobar. While the Descriptive research design is used in the analysis and determination of the need for a digitized system, the Developmental research design is used in the actual engineering and creation of the system, based on the identified functional and non-functional requirements. This is in line with the use and application of the Agile Scrum Methodology in creating a flexible system.

3.1.1 Population and Locale of the Study

The population of this study includes the key stakeholders who are affiliated and associated with D'Saints Grill and Restobar in San Carlos City, Pangasinan. The population is divided into four different user levels. The participants include one (1) Administrator who oversees the governance of the system, one (1) Event Manager who oversees the internal operations of the system, and an estimated one hundred (100) Service Providers or outsourcers, and an estimated one thousand (1,000) Clients who comprise the primary end-users of the system. The researchers utilized a purposive sampling method in choosing the

participants for the study. The reason for choosing this method is that the researchers were able to target the participants who have firsthand knowledge and experience about the existing manual process. The reason for choosing this method is that the requirements gathered will specifically address the needs of the participants while still abiding by the set time and resource constraints.

The specific locale of this study is D'Saints Grill and Restobar, located in San Carlos City, Pangasinan. As a notable restaurant in the area that offers integrated services for events and catering, D'Saints Grill is a strategic and relevant site for the implementation of the EVENTECH platform. The restaurant is located in a rapidly urbanizing area, where it is increasingly challenged to deal with complex logistical issues involving multiple service providers. This makes D'Saints Grill an ideal site for this study, where a digital solution is sought for modernizing conventional business processes and facilitating inter-organizational collaboration involving its outsourced service providers.

3.1.2 Data Instrumentation

An Interview Guide was used as the primary data collection method for this capstone project to gather in-depth insights and qualitative perspectives from the respondents regarding the platform's interface design, the efficiency of service providers, the accuracy of the booking engine, and the overall usability of the system. The interview questions and structure were specifically modified and customized to align with the unique operational requirements of D'Saints Grill and Restobar.

Document Analysis was employed to examine the existing records and internal documentation used by D'Saints Grill and Restobar, such as manual event reservation forms and outsourcer contact logs. This method allowed the developers to critically assess

the establishment's current workflows and service coordination practices that informed the system's functional requirements. According to Davie and Wyatt (2023), document analysis involves interpreting records as artifacts that reflect the values and procedures of an organization. Through this method, the development team identified the specific gaps in manual record-keeping and the logistical inefficiencies that necessitated a digital intervention.

Interviews were conducted with key stakeholders, including the administrator, event managers, and selected service providers, to gather insights into the complexities of the current event planning process. These interviews were structured, meaning they were guided by a consistent set of questions, which ensured the reliability of the gathered data while allowing for in-depth clarifications concerning the challenges faced in coordinating with a number of outsourcers. In their article, Sharma and Singh (2023) state that interviews are significant in gathering context-rich data, as they enable respondents to clarify complex operations and give a deeper understanding of the experience of the user.

Internet Sources were utilized to provide the researchers with technical documentation and best practices relevant to integrated marketplace platforms and event management systems. The sources included peer-reviewed journals and case studies that centered on Laravel implementation, relational database management, and web-based service outsourcing. According to Cilliers and Viljoen (2021), the use of the internet for data gathering allows for access to updated data in a more informed and comprehensive manner. Online scholarly databases such as Google Scholar and ResearchGate were consulted to ensure the credibility of the technical and academic references used in the development of EVENTECH.

3.2 Computer Hardware and Software Requirements

Table 3.2.1 presents the minimum hardware specifications required for the developers to build and test EVENTECH: An Integrated Event Management System and Service Outsourcing Platform for D'Saints Grill and Restobar in San Carlos City, Pangasinan. It details essential components such as the processor, RAM, storage, operating system, and network interface to ensure an optimal development environment for coding, system testing, and deployment.

Table 3.2.1 Minimum Computer Hardware and Software Requirements

Component	Specification
Processor	Intel i3 or higher
Memory (RAM)	Minimum of 4 GB
Storage	256 GB
Operating System	Windows 11
Internet Connection	Required for access full functionalities
Display	1366 x 768 or higher
Input Devices	Keyboard, Mouse, Touchpad
Browser	Google Chrome (latest version)

The listed hardware and software specifications ensure that the system operates efficiently during development and actual use. These requirements support essential tasks such as running local servers, executing Laravel commands, and managing database operations. Google Chrome is the recommended browser due to its compatibility and reliable performance with the modern web technologies implemented in the system.

3.3 Software Requirements for System Development

Table 3.3.1 presents the software tools and technologies used for developing the test EVENTECH: An Integrated Event Management System and Service Outsourcing

Platform. These tools were chosen for their reliability, ease of use, and suitability for rapid development and deployment.

Table 3.3.1 Software Requirements for System Development

Component	Specification
Operating System	Windows 11
Browser	Windows 11 Browser Google Chrome
Integrated Development Environment	Visual Studio Code
Terminal/Shell	PowerShell / Terminal
Programming Language	PHP
Front end Technologies	HTML, CSS, Bootstrap, JavaScript
Database Management System (DBMS)	MySQL
Framework	Laravel
Web Server	XAMPP

These software tools were selected to support structured and efficient system development. PHP and Laravel offer a reliable framework for building secure and scalable backend functionalities, while MySQL handles structured data storage and retrieval. Visual Studio Code provides a user-friendly yet powerful coding environment ideal for development tasks. XAMPP functions as a local server environment, allowing the system to be developed and tested under conditions that closely simulate actual deployment. This setup ensures consistency, reliability, and ease of testing throughout the development process.

3.4 System Framework

In this study, the system framework section outlines the specific procedures undertaken to gather and analyze information, serving as a structured guide for how the investigation was carried out.

The process of development of EVENTECH was facilitated by the Agile Scrum methodology, which ensured that there was constant communication among the team members. This would ensure that the organization was responsive to feedback from the stakeholders, D'Saints Grill and Restobar. The process was also divided into stages such as the generation of sprint backlogs, sprint planning, implementation, daily scrums, review, and retrospectives. The process ensured that the performance of the system was optimized at all times, hence ensuring that the final product would be responsive to the needs of the administrator, the managers of events, service outsourcers, and clients.

Scrum is an Agile method for managing a project that allows teams to manage their activities by employing a set of values, principles, and practices. It is a popular method in the field of software development and other related fields that help teams divide their tasks into a sprint or a set of time periods in which to effectively finish tasks. The main goal of a sprint is to create the most important features first, such as the service outsourcing marketplace and event coordination tools. Within no more than two to four weeks, a potentially functional deliverable product should be created. The combination of the Scrum method and the Agile philosophy results in the Agile Scrum method. It allows teams to effectively work on a project in small pieces. It is a lightweight method that offers just the right amount of structure for individuals and teams to use in their daily lives. It allows for cooperatively producing incremental value.

The use of the Agile Scrum framework was appropriate for this capstone project as it ensures a structured yet flexible approach to system development. It allowed the team to incorporate feedback regularly, quickly responded to changes, and incrementally built and improved the system, leading to a more effective and user-centered outcome.

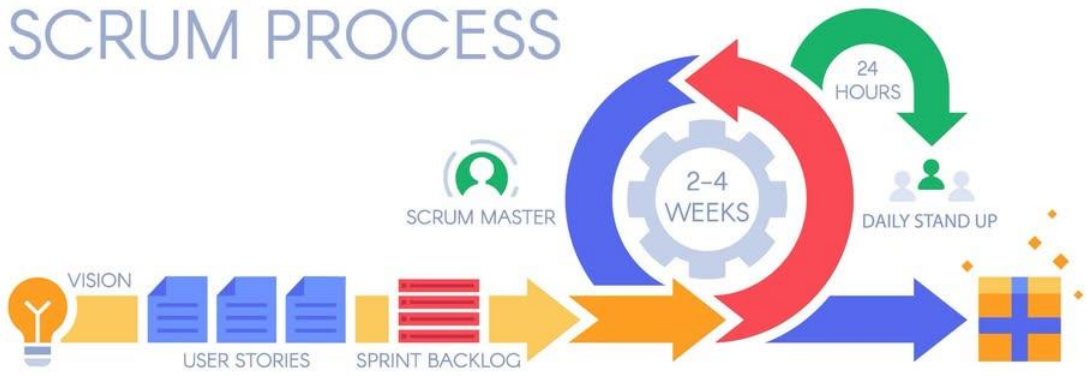


Figure 3.4.1 Agile Scrum Methodology

Figure 3.4.1 showcases the Agile Scrum Methodology, which is an adaptable software development methodology that emphasizes cooperation, adaptability, and the swift delivery of functional software components. The Agile Scrum methodology is defined by Sprints, which are software development cycles that are normally between two to four weeks long. During each sprint, a certain number of features for a platform are developed, tested, and reviewed. This software development methodology offers an opportunity for feedback and refinement for the entire software development process so that the final product meets the needs of the event management industry.

The developers chose to employ this methodology due to its adaptability to the changing requirements as well as the user-centered nature of the approach. In the case of EVENTECH, the users include the administrator, managers of the event, the service providers, as well as the clients. The complex functionalities that the software had to support included the outsourcing of services, the coordination of the event in real-time, as well as the management of D'Saints Grill and Restobar. The Agile Scrum approach allowed the development team to test the critical functionalities in increments.

Furthermore, as Agile Scrum's emphasis on communication and sprint reviews has shown, as discussed by Khan et al. (2023), this helps to greatly enhance transparency, accountability, and satisfaction during the development of the software. This ensures that feedback from the managers and the service providers is incorporated into the system quickly, thus creating a better-functioning, efficient, and user-oriented platform. This method was selected by the developers for a number of reasons, first being that it offers the necessary structure for managing the complexity of the project while ensuring high-quality results.

Phases of the Agile Scrum

The development of EVENTECH: An Integrated Event Management System and Service Outsourcing Platform followed the Agile Scrum methodology. Scrum is an iterative approach that emphasizes continuous delivery, stakeholder feedback, and team collaboration. The following outlines the phases based on the Scrum framework illustrated in the development cycle:

Product Backlog

The first step was the exhaustive collection of all the necessary features, technical requirements, and specifications for the system. The developers worked meticulously to identify the key requirements for the system, including the marketplace for service providers, automated event scheduling tools, and specific management tools for D'Saints Grill and Restobar. The requirements were then collated into a master list based on their criticality for the system's architecture. By studying the gaps in the manual process for event coordination, the developers ensured that the list was for a system that was not only innovative but also practical for the target users.

Sprint Planning

In the Sprint Planning phase of the project, the development team carried out strategic sessions to identify specific high-priority items to be executed within a fixed period of time or "Sprint." The team set specific goals to be achieved during the project iterations. The project development team focused on developing fundamental features such as user authentication, secure registration, and the interface for outsourcing the service. This phase of the project was essential in the allocation of resources and efficient time management. This is because the project development team managed to break down the complex requirements of the system into smaller tasks.

Sprint Implementation

The Implementation phase was the primary technical execution phase for the development of the EVENTECH platform, which was centered on the development of the web-based modules by means of rigorous coding and integration into the databases. For a professional and user-friendly interface for the platform, the development team opted to employ Bootstrap 5 and CSS for the development of the platform. This phase was marked by the development of key features for the platform, including the administrative dashboards, booking engine, and the logic for the outsourcing of services. This was a critical phase in the development of the platform since it ensured that the integration between the clients and the service providers was meticulously done so that the end product was a web-based platform that was successful in the objective of being a comprehensive all-in-one platform for event management.

Daily Scrum

Throughout the development lifecycle, the team maintained a series of brief synchronized meetings referred to as Daily Scrums. This provided a platform for the development team to discuss the progress being made on the respective web modules that had been allocated to them. It was also a platform for the development team to share information on any potential "blockers" or issues that could affect the development of the system. This was a key milestone in ensuring that the development of EVENTECH was on course. It ensured that the development team was agile enough to address any issues that arose from the development sprint.

Sprint Review

As a result, a Sprint Review meeting was conducted at the end of each sprint, where the newly developed functional web modules were presented to the stakeholders, which included the management of D'Saints Grill and Restobar, as well as academic advisers. This phase was geared toward ensuring that the "potentially shippable" product increment met the quality standards, as well as academic requirements, for each feature developed during the course of the sprint. The feedback gathered during these meetings proved useful for the team, as they were able to get a glimpse into how the system performed, hence allowing for necessary adjustments where needed before moving on to the next course of action.

Sprint Retrospective

After this review process, the team carried out a Sprint Retrospective meeting where they reflected on their internal development process. The developers reflected on their process of coding and the ways in which they collaborated in order to identify the effective ways of coding and collaboration that needed improvement. This process of self-

assessment was very important in optimizing their efficiency and improving their technical collaboration. This way, the developers would ensure that the integrated system of EVENTECH would be very efficient, effective, and tailored for the needs of the administrators, service providers, and clients through its web-based nature.

3.5 Tools for Data Analysis – Data Flow Diagram, Use Case, ERD

This section presents the tools used to analyze and visualize the system's processes. Models such as Data Flow Diagrams (DFD), Use Case Diagrams, and Entity-Relationship Diagrams (ERD) were utilized to provide a clear and structured representation of the system's workflow, functionalities, and data relationships. These tools helped in understanding how data moves within the system, how users 41 interacted with it, and how entities were related, ensuring a comprehensive view of the system's design and operation.

Data Flow Diagram

A Data Flow Diagram (DFD) is a visual tool used in system analysis and design to show how data moves through a system. It demonstrates how data enters the system, how it is processed, and how it exits as output. A DFD displays the sources and destinations of data, the locations where data is stored, and the pathways through which data flows within the system. It uses standard symbols such as rectangles to represent external entities or sources of data, circles or ovals to illustrate processes that transform the data, and open-ended rectangles to indicate data stores where information is kept. Arrows are used to show the direction of data movement between these elements. By presenting the flow of information clearly, DFDs help analysts and developers understand system operations, identify inefficiencies, and design more effective methods for processing and managing data.

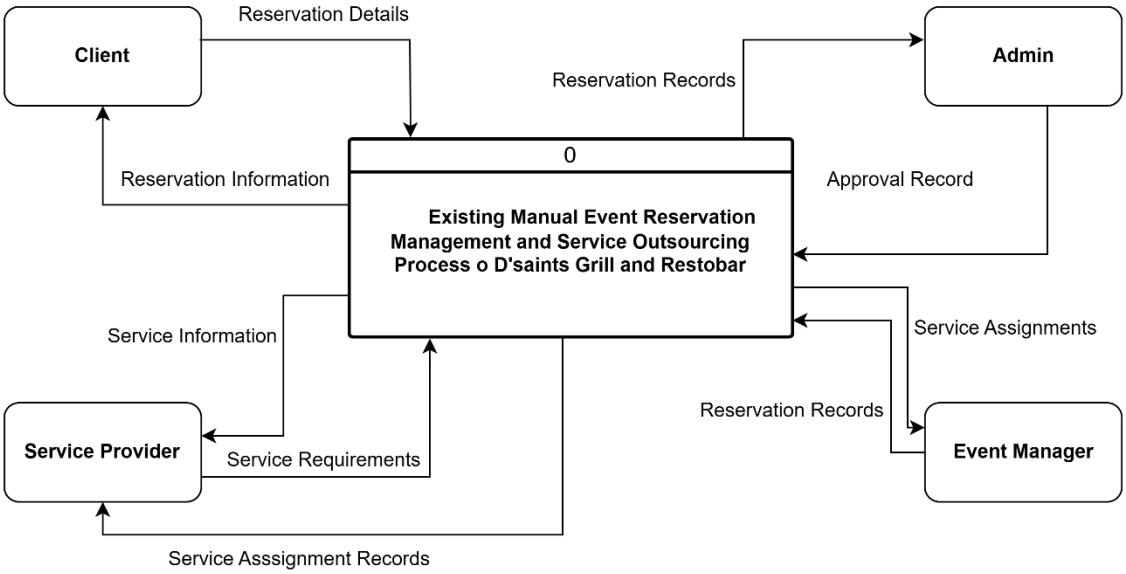


Figure 3.5.1 Level 0 Data Flow Diagram for existing Event Reservation Management and Service Outsourcing Platform for D'saints Grill and Restobar

Figure 3.5.1 illustrates the Level 0 Data Flow Diagram of the existing reservation management and service outsourcing process of D'saints Grill and Restobar. It presents how information is exchanged among the four main external entities: the Client, the Admin, the Event Manager, and the Service Provider in the current manual or semi-manual setup before the proposed system was developed.

The Client initiates the process by sending reservation inquiries, event details, and service requests to D'saints Grill and Restobar. These details may include the preferred date, time, number of guests, event type, and other specific needs. In the existing process, this information is usually communicated through face-to-face interaction, phone calls, or online messaging. After processing the request, the client receives reservation confirmation, event updates, and other service-related information.

The Admin is responsible for receiving and organizing reservation details, maintaining records, and monitoring transactions. In the current process, the admin

manually records client information, reservation schedules, and service requests. The admin also coordinates with the event manager and keeps track of the information needed for business operations. Because the process is handled manually, data retrieval and monitoring may take time and may be prone to errors or duplication.

The Event Manager handles the planning and coordination of reservations and outsourced services based on the client's requests. The event manager communicates with both the admin and service providers to arrange the necessary requirements for each event or reservation. In the existing process, schedules, service details, and event preparations are managed through manual coordination, which may cause delays and difficulty in updating information.

The Service Provider receives outsourced service requests and schedule details from the event manager or admin. They also provide updates regarding service availability, assigned tasks, and service status. Since the current process relies mainly on manual communication, the exchange of information between the restobar and service providers may not always be immediate and properly centralized.

Overall, the diagram presents a high-level view of the existing reservation management and service outsourcing process of D'saints Grill and Restobar. It highlights that the current workflow depends on manual recording, traditional communication methods, and separate coordination among involved entities. This existing process served as the basis for developing the proposed system shown in the next level of the data flow diagram.

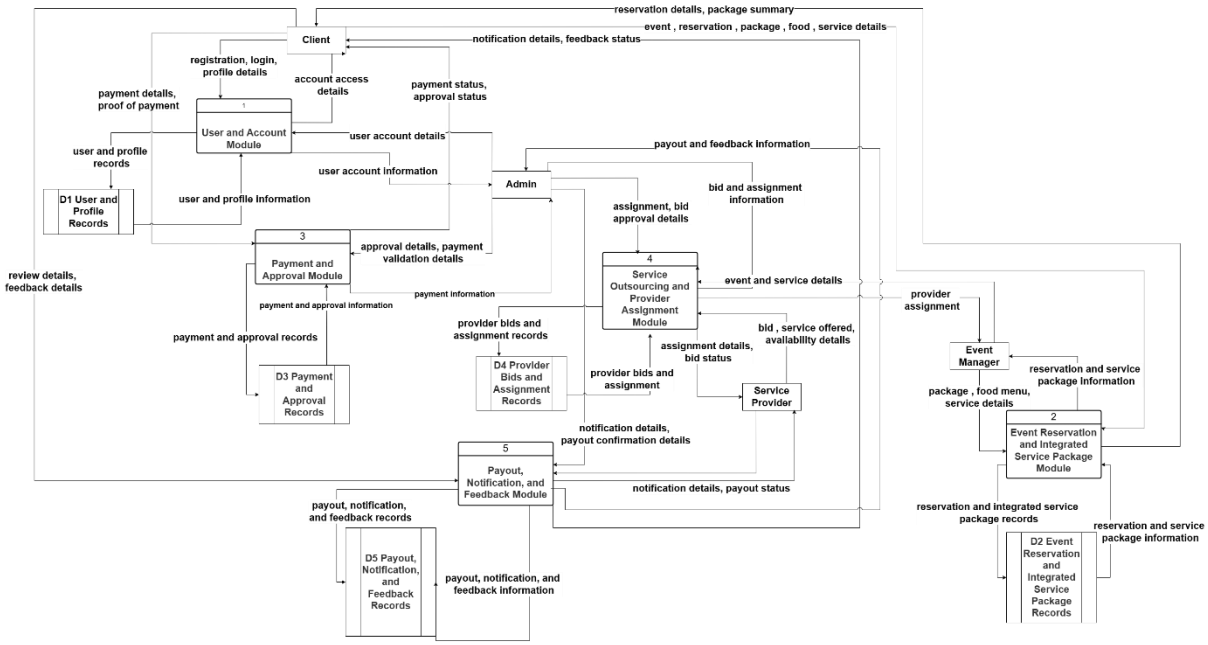


Figure 3.5.2 Level 1 Data Flow Diagram for the proposed Event Management System and Service Outsourcing Platform of D'saints Grill and Restobar

Figure 3.5.2 presents the Level 1 Data Flow Diagram of the proposed Event Management System and Service Outsourcing Platform of D'saints Grill and Restobar. It illustrates how the Client, Admin, Event Manager, and Service Provider interact with the system processes and how data is stored and retrieved through D1: User and Profile Records, D2: Event Reservation and Integrated Service Package Records, D3: Payment and Approval Records, D4: Provider Bids and Assignment Records, and D5: Payout, Notification, and Feedback Records.

The process begins with User and Account Module (1), where the client performs registration, login, and profile management. The user's account and profile information are stored in D1: User and Profile Records. This module ensures secure access and management of user information within the system.

In Event Reservation and Integrated Service Package Module (2), the client submits event details and selects packages, food, and services. The system processes these

selections and stores the reservation and integrated service package information in D2: Event Reservation and Integrated Service Package Records. The client then receives reservation details and package summary.

During Payment and Approval Module (3), the client submits payment details and proof of payment. The admin validates and approves or rejects the transaction. The payment and approval information are stored in D3: Payment and Approval Records, and the client is notified of the payment status.

The Service Outsourcing and Provider Assignment Module (4) handles the interaction between the event manager and service providers. Based on event requirements, service providers submit bids, service offers, and availability. These are recorded in D4: Provider Bids and Assignment Records. The system evaluates bids and assigns providers to the event.

Finally, the Payout, Notification, and Feedback Module (5) manages provider payouts, system notifications, and user feedback. Service providers receive payout confirmation, while notifications and feedback are handled and stored in D5: Payout, Notification, and Feedback Records.

Overall, the Level 1 Data Flow Diagram shows how the system integrates user management, event reservation, payment processing, service outsourcing, and payout handling into a centralized platform, improving efficiency, data organization, and communication among all users.

Use Case Diagram

A use case diagram is a visual representation that shows how users, known as actors, interact with a system to achieve specific goals. It provides a high-level overview

of the system's functionality by illustrating the different actions or services the system can perform and how users engage with them. In the given diagram, it clearly presents the interactions between clients, service providers, and administrators, including processes such as event creation, service selection, booking, payment processing, and system management. By organizing these interactions into a simple visual format, the diagram helps developers and stakeholders understand the system's scope, features, and user behavior more effectively. This makes it an essential tool during the planning and design phase, as it ensures that all participants have a clear and shared understanding of how the system will operate and how users will navigate through it (GeeksforGeeks, 2025).

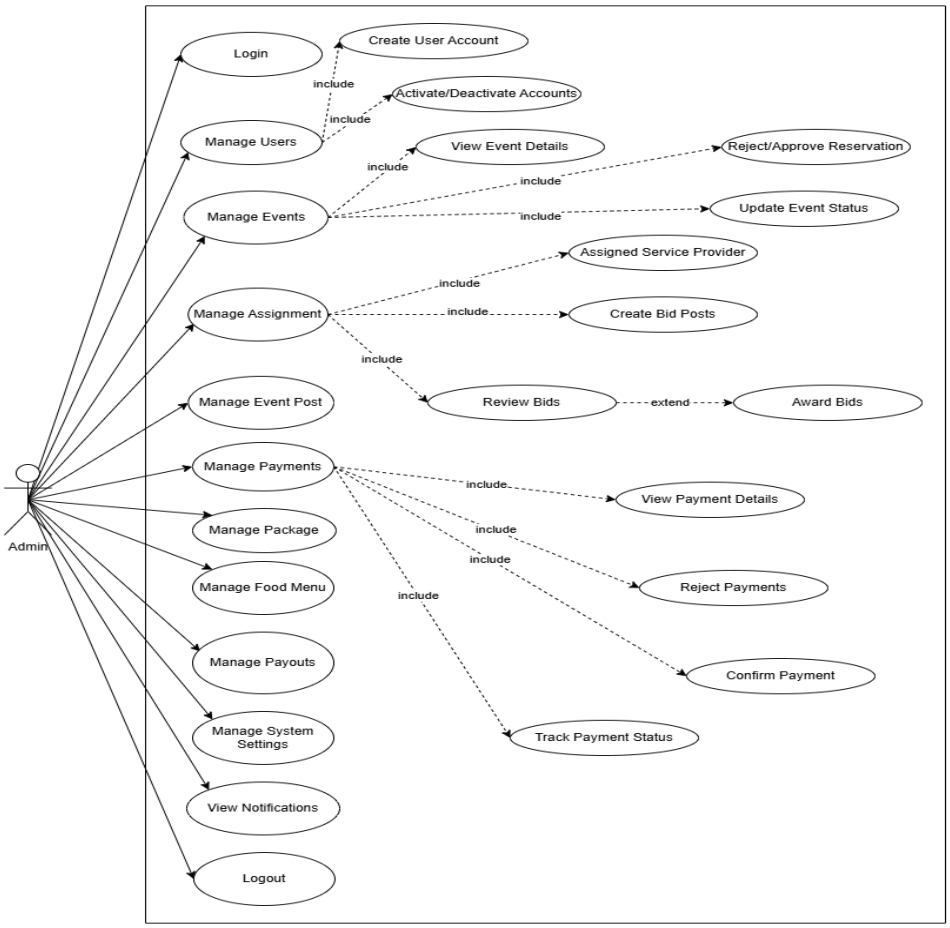


Figure 3.5.3 Shows the Use Case Diagram for the Admin for the An Integrated Event Management System and Service Outsourcing Platform

Figure 3.5.3 presents the Use Case Diagram showing the key functions performed by the Administrator in EVENTECH: An Integrated Event Management System and Service Outsourcing Platform. The administrator is responsible for managing users, events, assignments, event posts, packages, food menus, payouts, payments, and system settings. This includes creating user accounts, controlling user access, viewing event details, approving or rejecting reservations, updating event status, assigning service providers, creating bid posts, reviewing bids, and awarding bids. The administrator can also monitor payments and notifications to support secure, organized, and efficient system operations.

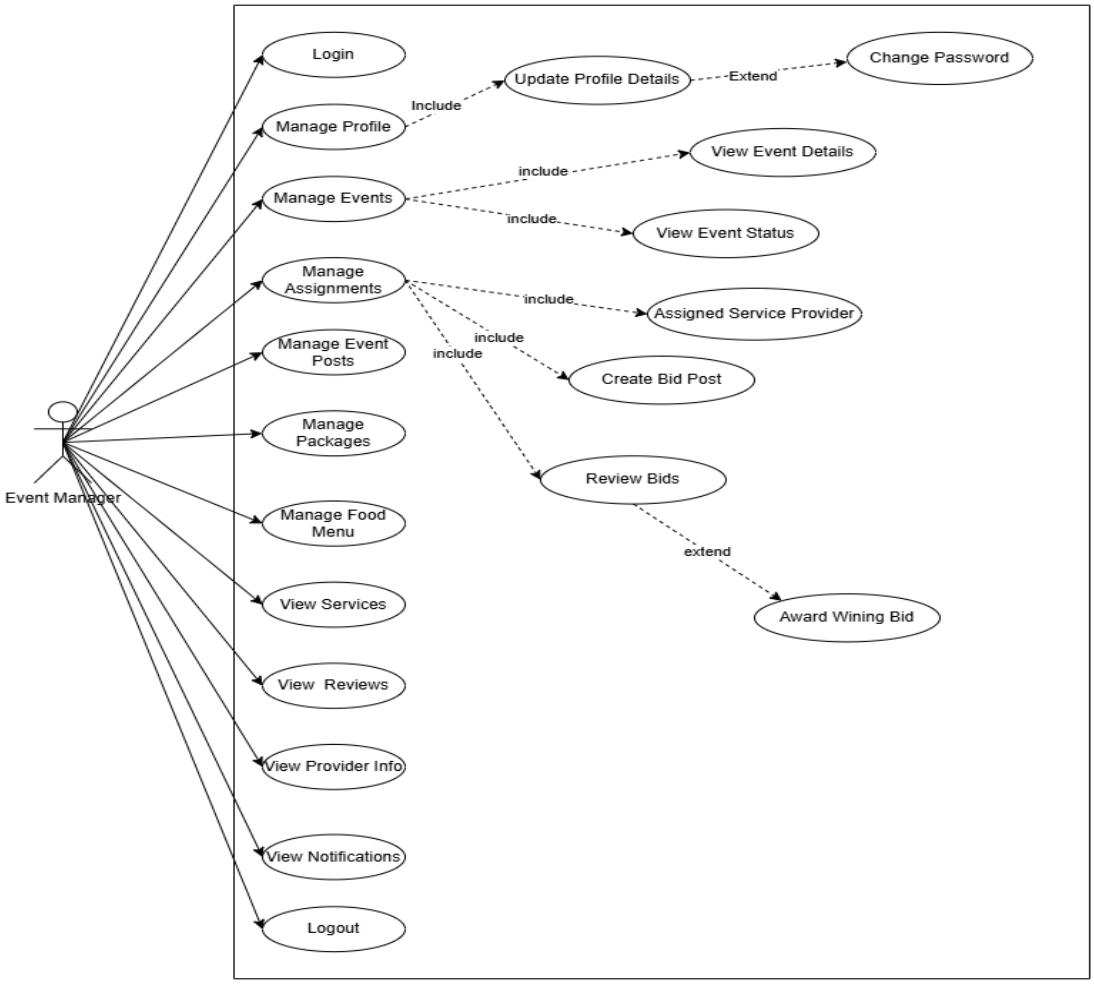


Figure 3.5.4 shows the Use Case Diagram for the Event Manager of the An Integrated Event Management System and Service Outsourcing Platform

Figure 3.5.4 shows the Use Case Diagram outlining the key functions of the Event Manager within the Event Management System and Service Outsourcing Platform. The Event Manager is responsible for managing events, assignments, event posts, packages, and food menu records. This includes viewing event details and status, assigning service providers, creating bid posts, reviewing submitted bids, and awarding the winning bid. The Event Manager can also manage profile details, change the account password, and view services, reviews, provider information, and notifications to support efficient event operations and provider coordination.

Figure 3.5.5 shows the Use Case Diagram for the Service Provider of the An Integrated Event Management System and Service Outsourcing System

Figure 3.5.5 shows the Use Case Diagram outlining the different actions a Service Provider can perform within the Event Management System and Service Outsourcing Platform. The service provider is responsible for managing profile details, services, bids, assigned events, payouts, and portfolio records. This includes registering or logging into the system, updating profile information, adding and updating services, placing or withdrawing bids, viewing assigned events, and managing payout requests. The service provider can also view notifications and log out after completing system activities. Through these functions, the service provider is able to participate in bidding, deliver services, and monitor payouts within the system.

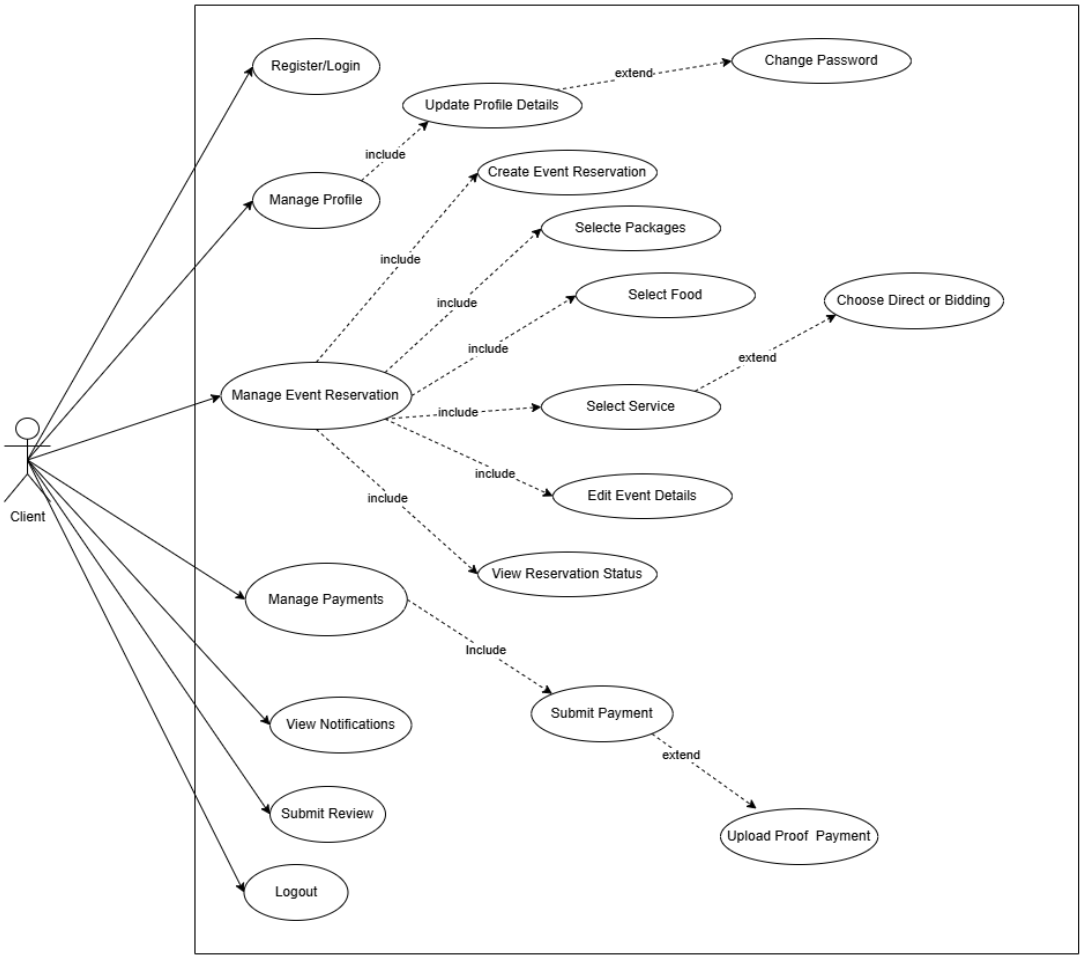


Figure 3.5.6 shows the Use Case Diagram for the Service Provider of the An Integrated Event Management System and Service Outsourcing System

Figure 3.5.6 shows the Use Case Diagram outlining the different actions a Client can perform within the Event Management System and Service Outsourcing Platform. The client is responsible for managing profile information, event reservations, payments, notifications, and reviews. This includes registering or logging into the system, updating profile details, changing the account password, creating event reservations, selecting packages, food, and services, and choosing whether a service will be direct or for bidding. Additionally, the client can edit event details, view reservation status, submit payments, upload proof of payment, view notifications, submit reviews, and log out after completing system activities. Through these functions, the client is able to reserve events, customize service requirements, monitor reservation progress, and complete payment transactions within the system.

Entity Relationship Diagram

An Entity-Relationship Diagram (ERD) is a graphical representation used in database design to show the entities in a system and the relationships among them. It serves as a structural model that identifies the major data components of the database, such as users, events, reservations, packages, food items, service providers, bids, payments, notifications, reviews, and system settings. In the EVENTECH system, the ERD is essential for illustrating how these entities are connected to support event management and service outsourcing operations. It helps developers and researchers understand the database structure, maintain data consistency, reduce design errors, and ensure that the system database is properly organized and reliable throughout the implementation of the project.

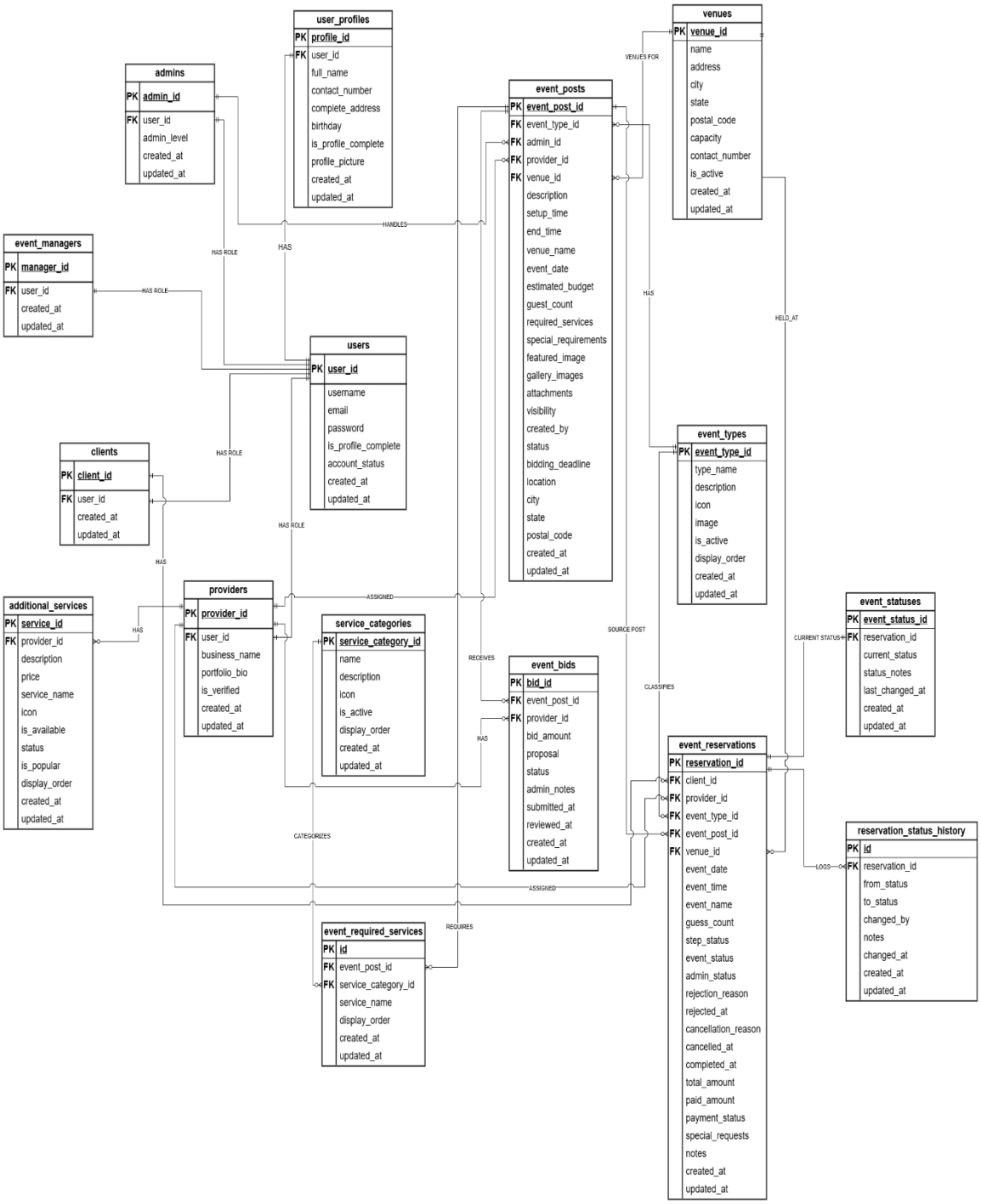


Figure 3.5.7. Entity Relationship Diagram of EVENTECH for Core System Modules

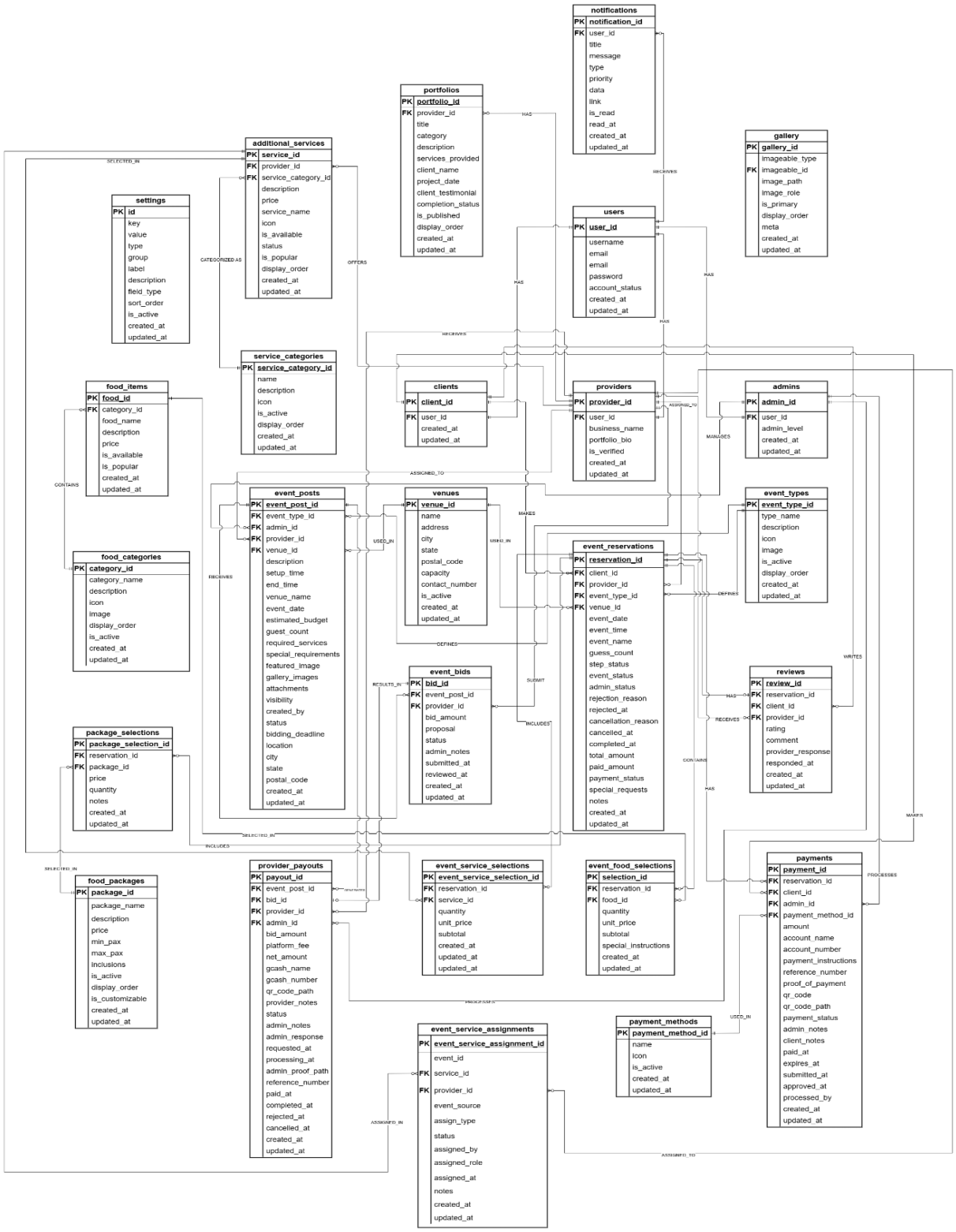


Figure 3.5.8. Entity Relationship Diagram of EVENTECH for Operational and Support Module

Data Dictionary

A Data Dictionary is a structured reference used in database design to describe and organize all data elements within the system. It identifies each field in every table, such as event details, user information, reservations, services, payments, and other related records, and specifies its data type, size, description, and purpose. By clearly defining these properties, the data dictionary standardizes how data is named, stored, and interpreted across the system.

This is essential in the development of the system because processes such as event management, reservation handling, service selection, payment processing, and user management depend on consistent and well-defined data fields. Having a structured data dictionary ensures that all information entered into the system is accurate, organized, and reliable.

The data dictionary also serves as a guide for developers, database designers, and system users by providing a clear understanding of how each field is used, how tables are related, and how data can be retrieved, processed, and analyzed efficiently. It helps reduce design errors, prevents inconsistencies, and improves the overall integrity and quality of the system.

Table 3.5.1 represents the Additional Services and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each service and stores information such as service name, description, pricing, availability, and status.

Table 3.5.1 additional_services

Fields	Data Type	Null	Keys	Description

service_id	bigint(20)	No		Service Id
user_id	bigint(20)	Yes		User Id
service_name	varchar(200)	No		Service Name
description	text	Yes		Description
price	decimal(10,2)	No		Price
service_category_id	bigint(20)	Yes		Service Category Id
icon	varchar(50)	Yes		Icon
is_available	tinyint(1)	No		Is Available
status	enum('active','inactive')	No		Status
is_popular	tinyint(1)	No		Is Popular
display_order	int(11)	No		Display Order
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.2 represents the Admins and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each administrator and stores information such as user reference and administrative level.

Table 3.5.2 admins

Fields	Data Type	Null	Keys	Description
admin_id	bigint(20)	No		Admin Id
user_id	bigint(20)	No		User Id

admin_level	varchar(255)	No		Admin Level
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.3 represents the Clients and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each client and stores information linking users to client roles.

Table 3.5.3 clients

Fields	Data Type	Null	Keys	Description
client_id	bigint(20)	No		Client Id
user_id	bigint(20)	No		User Id
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.4 represents the Event Bids and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each bid and stores information such as event reference, bidder, bid amount, proposal, and status.

Table 3.5.4 Event_bids

Fields	Data Type	Null	Keys	Description
bid_id	bigint(20)	No		Bid Id
event_id	bigint(20)	No		Event Id
user_id	bigint(20)	No		User Id

bid_amount	decimal(10,2)	No		Bid Amount
proposal	text	Yes		Proposal
status	enum('pending','accepted', 'rejected','withdrawn')	No		Status
admin_notes	text	Yes		Admin Notes
submitted_at	timestamp	No		Submitted At
reviewed_at	timestamp	Yes		Reviewed At
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.5 represents the Event Food Selections and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each selection and stores information such as reservation reference, food items, quantity, pricing, and special instructions.

Table 3.5.5 event_food_selections

Fields	**Data Type**	**Null**	**Keys**	**Description**
selection_id	bigint(20)	No		Selection Id
reservation_id	bigint(20)	No		Reservation Id
food_id	bigint(20)	No		Food Id
quantity	int(11)	No		Quantity
unit_price	decimal(10,2)	No		Unit Price

subtotal	decimal(10,2)	No		Subtotal
special_instructions	text	Yes		Special Instructions
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.6 represents the Event Managers and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each manager and stores information linking users assigned as event managers.

Table 3.5.6 event_managers

Fields	Data Type	Null	Keys	Description
manager_id	bigint(20)	No		Manager Id
user_id	bigint(20)	No		User Id
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.7 represents the Event Posts and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each event post and stores detailed information such as event type, schedule, venue, budget, requirements, visibility, and status.

Table 3.5.7 Events_posts

Fields	Data Type	Null	Keys	Description
id	bigint(20)	No		Id

title	varchar(255)	No		Title
event_type	varchar(100)	No		Event Type
description	text	Yes		Description
setup_time	time	Yes		Setup Time
end_time	time	Yes		End Time
venue_id	bigint(20)	Yes		Venue Id
event_date	date	Yes		Event Date
estimated_budget	decimal(10,2)	Yes		Estimated Budget
guest_count	int(11)	Yes		Guest Count
special_requirements	text	Yes		Special Requirements
visibility	varchar(20)	No		Visibility
created_by	bigint(20)	Yes		Created By
status	enum(‘draft’,’open’, ‘closed’,’cancelled’, ’awarded’)	No		Status
bidding_deadline	timestamp	Yes		Bidding Deadline
user_id	bigint(20)	Yes		User Id
provider_id	bigint(20)	Yes		Provider Id
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

location	varchar(255)	Yes		Location
city	varchar(100)	Yes		City
state	varchar(100)	Yes		State
postal_code	varchar(20)	Yes		Postal Code

Table 3.5.8 represents the Event Required Services and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each required service and stores information about services needed for specific event posts.

Table 3.5.8 event_required_services

Fields	Data Type	Null	Keys	Description
id	bigint(20)	No		Id
event_post_id	bigint(20)	No		Event Post Id
service_category_id	bigint(20)	Yes		Service Category Id
service_name	varchar(150)	Yes		Service Name
display_order	int(10)	No		Display Order
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.9 represents the Event Reservations and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for

each reservation and stores comprehensive details such as client and provider information, event schedule, guest count, status tracking, payment details, and special requests.

Table 3.5.9 event_reservations

Fields	Data Type	Null	Keys	Description
reservation_id	bigint(20)	No		Reservation Id
client_id	bigint(20)	No		Client Id
provider_id	bigint(20)	Yes		Provider Id
event_type_id	bigint(20)	No		Event Type Id
event_name	varchar(200)	Yes		Event Name
event_date	date	No		Event Date
event_time	time	No		Event Time
venue_id	bigint(20)	Yes		Venue Id
guest_count	int(11)	No		Guest Count
step_status	enum(‘Schedule’, ’Package_Selection’, ‘Food_Selection’, ‘Additional_Services’, ‘Confirmation’,’Completed’, ‘Cancalled’).’	No		Step Status
event_status	varchar(50)	Yes		Event Status
admin_status	varchar(255)	No		Admin Status

rejection_reason	text	Yes		Rejection Reason
rejected_at	timestamp	Yes		Rejected At
cancellation_reason	text	Yes		Cancellation Reason
cancelled_at	timestamp	Yes		Cancelled At
completed_at	timestamp	Yes		Completed At
total_amount	decimal(10,2)	No		Total Amount
paid_amount	decimal(10,2)	No		Paid Amount
payment_status	enum(‘Pending’,’Partial’, ‘Paid’,’Refunded’)	No		Payment Status
special_requests	text	Yes		Special Requests
notes	text	Yes		Notes
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.3 represents the Event Service Assignments and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each assignment and stores information such as the event source, service provider, assignment method (direct or bidding), and the current operational status of the service fulfillment.

Table 3.5.10 event_services_assignments

Fields	Data Type	Null	Keys	Description

id	bigint(20) unsigned	No	PRI	Unique primary identifier (Auto-increment).
event_source	enum('reservation', 'event_post')	No		Determines if the event is a direct reservation or a public post.
event_id	bigint(20) unsigned	No	MUL	The ID of the event from the source specified above.
service_id	bigint(20) unsigned	Yes	FK	Foreign key to additional_services. Nullable for bidding.
provider_id	bigint(20) unsigned	No	FK	Foreign key to users (the service provider).
assignment_type	enum('direct', 'bidding')	No	MUL	Defines how the provider was selected.
status	enum('pending', 'assigned', 'ongoing', 'completed')	No		Tracks the progress of the service fulfillment.
assigned_by	bigint(20) unsigned	Yes	FK	Foreign key to users (the admin/manager who assigned).
assigned_role	enum('admin', 'event_manager')	Yes		The authority level of the person who made the assignment.
assigned_at	timestamp	Yes		Date and time when the assignment was finalized.
notes	text	Yes		Specific details (e.g., bid amounts or client names).

created_at	timestamp	Yes		Record creation timestamp.
updated_at	timestamp	Yes		Record update timestamp.

Table 3.5.11 represents the Event Service Selections and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each service selection and stores information such as selected services, quantity, pricing, and subtotal for each reservation.

Table 3.5.11 event_service_selections

Fields	Data Type	Null	Keys	Description
id	bigint(20)	No		Id
reservation_id	bigint(20)	No		Reservation Id
service_id	bigint(20)	No		Service Id
quantity	int(11)	No		Quantity
unit_price	decimal(10,2)	No		Unit Price
subtotal	decimal(10,2)	No		Subtotal
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.12 represents the Event Statuses and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each event status and stores information such as the current status, status notes, and timestamps of status changes for each reservation.

Table 3.5.12 event_statuses

Fields	Data Type	Null	Keys	Description
event_status_id	bigint(20)	No		Event Status Id
reservation_id	bigint(20)	No		Reservation Id
current_status	varchar(50)	No		Current Status
status_notes	text	Yes		Status Notes
last_changed_at	timestamp	Yes		Last Changed At
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.13 represents the Event Types and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each event type and stores information such as type name, description, visual elements, and display configuration.

Table 3.5.13 event_types

Fields	Data Type	Null	Keys	Description
event_type_id	bigint(20)	No		Event Type Id
type_name	varchar(100)	No		Type Name
description	text	Yes		Description
icon	varchar(50)	No		Icon
image	varchar(255)	Yes		Image

is_active	tinyint(1)	No		Is Active
display_order	int(11)	No		Display Order
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.14 represents the Food Categories and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each food category and stores information such as category name, description, images, and display settings.

Table 3.5.14 food_categories

Fields	Data Type	Null	Keys	Description
category_id	bigint(20)	No		Category Id
category_name	varchar(100)	No		Category Name
description	text	Yes		Description
icon	varchar(50)	Yes		Icon
image	varchar(255)	Yes		Image
display_order	int(11)	No		Display Order
is_active	tinyint(1)	No		Is Active
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.15 represents the Food Items and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each food item and stores detailed information such as category reference, food name, pricing, ingredients, nutrition information, availability, and preparation details.

Table 3.5.15 food_items

Fields	Data Type	Null	Keys	Description
food_id	bigint(20)	No		Food Id
category_id	bigint(20)	No		Category Id
food_name	varchar(200)	No		Food Name
description	text	Yes		Description
price	decimal(10,2)	No		Price
is_available	tinyint(1)	No		Is Available
is_popular	tinyint(1)	No		Is Popular
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.16 represents the Food Packages and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each food package and stores information such as package name, description, pricing, guest capacity, inclusions, and customization options.

Table 3.5.16 food_packages

Fields	Data Type	Null	Keys	Description

package_id	bigint(20)	No		Package Id
package_name	varchar(200)	No		Package Name
description	text	Yes		Description
price	decimal(10,2)	No		Price
min_pax	int(11)	No		Min Pax
max_pax	int(11)	Yes		Max Pax
inclusions	longtext	Yes		Inclusions
is_active	tinyint(1)	No		Is Active
display_order	int(11)	No		Display Order
is_customizable	tinyint(1)	No		Is Customizable
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.17 represents the Gallery and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each image and stores information such as image type, associated record, file path, role, and display configuration.

Table 3.5.17 gallery

Fields	Data Type	Null	Keys	Description
gallery_id	bigint(20)	No		Gallery Id
imageable_type	varchar(150)	No		Imageable Type

imageable_id	bigint(20)	No		Imageable Id
image_path	varchar(255)	No		Image Path
image_role	varchar(50)	No		Image Role
is_primary	tinyint(1)	No		Is Primary
display_order	int(10)	No		Display Order
meta	longtext	Yes		Meta
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.18 represents the Notifications and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each notification and stores information such as user reference, message content, type, priority, status, and timestamps.

Table 3.5.18 notifications

Fields	Data Type	Null	Keys	Description
notification_id	bigint(20)	No		Notification Id
user_id	bigint(20)	No		User Id
title	varchar(255)	No		Title
message	text	No		Message
type	varchar(255)	No		Type
priority	varchar(255)	No		Priority

data	longtext	Yes		Data
link	varchar(255)	Yes		Link
is_read	tinyint(1)	No		Is Read
read_at	timestamp	Yes		Read At
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.19 represents the Package Selections and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each package selection and stores information such as reservation reference, selected package, pricing, quantity, and notes.

Table 3.5.19 package_selections

Fields	Data Type	Null	Keys	Description
package_selection_id	bigint(20)	No		Package Selection Id
reservation_id	bigint(20)	No		Reservation Id
package_id	bigint(20)	No		Package Id
price	decimal(10,2)	No		Price
quantity	int(11)	No		Quantity

Table 3.5.20 represents the Payment Methods and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for

each payment method and stores information such as method name, icon, and availability status.

Table 3.5.20 payment_methods

Fields	Data Type	Null	Keys	Description
payment_method_id	bigint(20)	No		Payment Method Id
name	varchar(100)	No		Name
icon	varchar(100)	Yes		Icon
is_active	tinyint(1)	No		Is Active
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.21 represents the Payments and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each payment and stores detailed information such as reservation reference, payment amount, method, account details, transaction status, and timestamps.

Table 3.5.21 payments

Fields	Data Type	Null	Keys	Description
payment_id	bigint(20)	No		Payment Id
reservation_id	bigint(20)	No		Reservation Id
amount	decimal(10,2)	No		Amount
payment_method_id	bigint(20)	Yes		Payment Method Id

account_name	varchar(255)	Yes		Account Name
account_number	varchar(255)	Yes		Account Number
reference_number	varchar(255)	Yes		Reference Number
proof_of_payment	varchar(255)	Yes		Proof Of Payment
qr_code	varchar(255)	Yes		Qr Code
payment_status	enum(‘pending’,’paid’, ‘rejected’,’cancelled’, ‘waiting_confirmation’)	No		Payment Status
admin_notes	text	Yes		Admin Notes
client_notes	text	Yes		Client Notes
paid_at	timestamp	Yes		Paid At
processed_by	bigint(20)	Yes		Processed By
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.22 represents the Portfolios and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each portfolio and stores information such as title, category, project details, services provided, client information, and publication status.

Table 3.5.22 portfolios

Fields	Data Type	Null	Keys	Description

portfolio_id	bigint(20)	No		Portfolio Id
user_id	bigint(20)	No		User Id
title	varchar(255)	No		Title
category	varchar(100)	Yes		Category
description	text	Yes		Description
services_provided	longtext	Yes		Services Provided
client_name	varchar(255)	Yes		Client Name
project_date	date	Yes		Project Date
client_testimonial	text	Yes		Client Testimonial
completion_status	varchar(255)	No		Completion Status
is_published	tinyint(1)	No		Is Published
display_order	int(11)	No		Display Order
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.23 represents the Provider Payouts and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each payout and stores detailed financial information such as bid amount, platform fee, net amount, payment details, status, and transaction records.

Table 3.5.23 provider_payouts

Fields	Data Type	Null	Keys	Description

payout_id	bigint(20)	No		Payout Id
event_id	bigint(20)	No		Event Id
bid_id	bigint(20)	No		Bid Id
provider_id	bigint(20)	No		Provider Id
admin_id	bigint(20)	Yes		Admin Id
bid_amount	decimal(10,2)	No		Bid Amount
platform_fee	decimal(10,2)	No		Platform Fee
net_amount	decimal(10,2)	No		Net Amount
gcash_name	varchar(255)	No		Gcash Name
gcash_number	varchar(255)	No		Gcash Number
qr_code_path	varchar(255)	Yes		Qr Code Path
provider_notes	text	Yes		Provider Notes
status	enum(‘pending’,’processing’, ‘paid’,’completed’,’rejected’, ‘cancelled’)	No		Status
admin_notes	text	Yes		Admin Notes
admin_response	text	Yes		Admin Response
requested_at	timestamp	Yes		Requested At
processing_at	timestamp	Yes		Processing At
admin_proof_path	varchar(255)	Yes		Admin Proof Path

reference_number	varchar(100)	Yes		Reference Number
paid_at	timestamp	Yes		Paid At
completed_at	timestamp	Yes		Completed At
rejected_at	timestamp	Yes		Rejected At
cancelled_at	timestamp	Yes		Cancelled At
transaction_reference	varchar(255)	Yes		Transaction Reference
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.24 represents the Providers and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each provider and stores information such as user reference, business name, portfolio details, and verification status.

Table 3.5.24 providers

Fields	Data Type	Null	Keys	Description
provider_id	bigint(20)	No		Provider Id
user_id	bigint(20)	No		User Id
business_name	varchar(255)	Yes		Business Name
portfolio_bio	text	Yes		Portfolio Bio
is_verified	tinyint(1)	No		Is Verified
created_at	timestamp	Yes		Created At

updated_at	timestamp	Yes		Updated At

Table 3.5.25 represents the Reservation Status History and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each record and stores information about status transitions, including previous and current status, responsible user, and timestamps.

Table 3.5.25 reservation_status_history

Fields	Data Type	Null	Keys	Description
id	bigint(20)	No		Id
reservation_id	bigint(20)	No		Reservation Id
from_status	varchar(50)	Yes		From Status
to_status	varchar(50)	No		To Status
changed_by	bigint(20)	Yes		Changed By
notes	text	Yes		Notes
changed_at	timestamp	Yes		Changed At
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.26 represents the Reviews and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each review and stores information such as reservation reference, client and provider details, ratings, feedback, and response records.

Table 3.5.26 reviews

Fields	Data Type	Null	Keys	Description
review_id	bigint(20)	No		Review Id
reservation_id	bigint(20)	No		Reservation Id
client_id	bigint(20)	No		Client Id
provider_id	bigint(20)	Yes		Provider Id
rating	tinyint(3)	No		Rating
comment	text	Yes		Comment
provider_response	text	Yes		Provider Response
responded_at	timestamp	Yes		Responded At
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.27 represents the Service Categories and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each service category and stores information such as name, description, icon, status, and display configuration.

Table 3.5.27 service_categories

Fields	Data Type	Null	Keys	Description
service_category_id	bigint(20)	No		Service Category Id
name	varchar(100)	No		Name

description	text	Yes		Description
icon	varchar(100)	Yes		Icon
is_active	tinyint(1)	No		Is Active
display_order	int(10)	No		Display Order
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.28 represents the Settings and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each setting and stores configuration details such as key-value pairs, type, grouping, labels, and system preferences.

Table 3.5.28 settings

Fields	Data Type	Null	Keys	Description
id	bigint(20)	No		Id
key	varchar(255)	No		Key
value	text	Yes		Value
type	varchar(255)	No		Type
group	varchar(255)	No		Group
label	varchar(255)	Yes		Label
description	text	Yes		Description
field_type	varchar(255)	No		Field Type

sort_order	int(11)	No		Sort Order
is_active	tinyint(1)	No		Is Active
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.29 represents the User Profiles and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each profile and stores personal information such as full name, contact details, address, birthday, and profile completion status.

Table 3.5.29 user_profiles

Fields	Data Type	Null	Keys	Description
profile_id	bigint(20)	No		Profile Id
user_id	bigint(20)	No		User Id
full_name	varchar(100)	Yes		Full Name
contact_number	varchar(15)	Yes		Contact Number
complete_address	text	Yes		Complete Address
birthday	date	Yes		Birthday
is_profile_complete	tinyint(1)	No		Is Profile Complete
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.30 represents the Users and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each user and stores essential account information such as username, email, password, and account status.

Table 3.5.30 users

Fields	Data Type	Null	Keys	Description
user_id	bigint(20)	No		User Id
username	varchar(50)	No		Username
email	varchar(100)	No		Email
password	varchar(255)	No		Password
is_profile_complete	tinyint(1)	No		Is Profile Complete
account_status	varchar(20)	No		Account Status
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

Table 3.5.31 represents the Venues and describes the data type and size that the proponents use. It contains a unique field that serves as the primary identifier for each venue and stores information such as name, location, capacity, contact details, and availability status.

Table 3.5.31 venues

Fields	Data Type	Null	Keys	Description
venue_id	bigint(20)	No		Venue Id

name	varchar(255)	No		Name
address	varchar(255)	Yes		Address
city	varchar(100)	Yes		City
state	varchar(100)	Yes		State
postal_code	varchar(20)	Yes		Postal Code
capacity	int(10)	Yes		Capacity
contact_number	varchar(50)	Yes		Contact Number
is_active	tinyint(1)	No		Is Active
created_at	timestamp	Yes		Created At
updated_at	timestamp	Yes		Updated At

3.6 The Proposed Implementation Plan

In this section, the developers present the proposed implementation plan for An Integrated Event Management System and Service Outsourcing Platform. This plan ensures a structured and organized deployment of the system, covering each phase from preparation to full implementation.

The implementation of the system aims to improve the process of event planning, service coordination, and outsourcing management through a centralized digital platform. It provides an automated and efficient solution for clients, service providers, event managers, and administrators by integrating features such as event posting, service bidding, reservation management, package and food selection, payment processing, notifications, and analytics dashboards.

Before the development and deployment of the system, the developers conducted planning, interviews, and requirement-gathering activities to identify the common challenges experienced in traditional event planning and manual service coordination. The information gathered helped determine the system's major functionalities, including user registration, event creation, provider bidding, service assignment, payment monitoring, and report generation.

The development of the system followed the Agile Scrum methodology, which allowed the developers to implement the platform in short and iterative development cycles. Each sprint focused on a particular module, such as user management, event posting, bidding, reservation handling, payment processing, and reporting. Feedback collected during each phase was used to improve the usability, efficiency, and overall performance of the system.

Once the system reached functional completeness and passed internal testing, the developers prepared it for pilot implementation. This phase included installing and configuring the system, setting up the database, preparing user accounts, and ensuring that all required modules were operational.

Following installation, an orientation session was conducted for the target users. This included demonstrations of the major workflows of the system, such as event posting, bid submission, service selection, reservation confirmation, and payment tracking. User guides and training materials were also prepared to help users understand the system more effectively.

A user training program was then conducted for the different stakeholders of the platform. Clients were trained on how to create event requests and manage bookings,

service providers were guided on how to submit bids and update service offerings, event managers were instructed on how to coordinate event requirements, and administrators were oriented on monitoring system activities and generating reports. The goal of this training was to ensure that all users could operate the system efficiently and independently. During the initial launch of the system, the developers remained available to provide technical support and address any issues encountered by the users. Feedback gathered during this stage was used to make necessary adjustments, enhance system performance, and improve the overall user experience.

By following this implementation plan, the developers aim to deliver a reliable, efficient, and user-friendly platform that enhances event management processes and improves service outsourcing coordination through the use of digital technology.

Strategies	**Activities**	**Personnel Involved**	**Duration**
Approval from D'Saints Grill and Restobar	Submission of the proposal and formal request for system implementation approval	Project Group Members, Event Manager/Administrator	1 Day
System Installation and Setup	Deployment of the system, database setup, software installation, and configuration of user access	Project Group Members, Assigned Technical Personnel	1 Day
Intro Session Materials	Preparation of printed guides, tutorials, and walkthrough presentations	Project Group Members	8 hours

User Training and Onboarding	Conducting training sessions for clients, providers, event managers, and administrators	Project Group Member, System End-users	5 days

Table 3.6.1 Implementation Plan

3.7 Statistical Tools

Scales and Points of the Questionnaire Survey

The proponents calculated the general mean for data treatment, and the findings were evaluated using the Likert Scale. Each category consisted of a linear scale that rates from 5 as very satisfied, 4 as satisfied, 3 as neutral, 2 as dissatisfied and 1 as very dissatisfied

Table 3.7.1 Verbal Interpretation for User Acceptability Testing

Numerical Equivalent	Weighted Value	Descriptive Equivalent	Descriptive Interpretation
5	4.51 - 5.00	Highly Acceptable	Strongly Agree
4	3.51 - 4.50	Acceptable	Agree
3	2.51 - 3.50	Moderately Acceptable	Moderately Agree
2	1.51 - 2.50	Fairly Acceptable	Disagree
1	1.00 - 1.50	Poorly Acceptable	Strongly Disagree

Scores ranging from 4.51 to 5.0 were viewed as reflective of a strong level of acceptability, implying that the system has effectively fulfilled users' needs and anticipations. Conversely, scores between 1.00 and 1.50 were categorized as reflecting low

acceptability, indicating user discontent and emphasizing that the system did not fulfill user requirements. Thorough examination of these diminished scores was vital to pinpoint particular areas needing enhancement and to guarantee the system more closely matches user expectations and preferences. This evaluation framework was used to measure user satisfaction and to guide system enhancements aimed at achieving optimal performance and user experience.

Range of the Weighted Mean and its Interpretation

The sampling technique utilized for testing the user acceptability was random sampling. Further, the average weighted mean was used in measuring the level of acceptability in terms of Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability and Security, Maintainability, and Flexibility

where:

$$\overline{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

x_i represents each value in the dataset

w_i represents the weight corresponding to each value

n is the number of values in the dataset

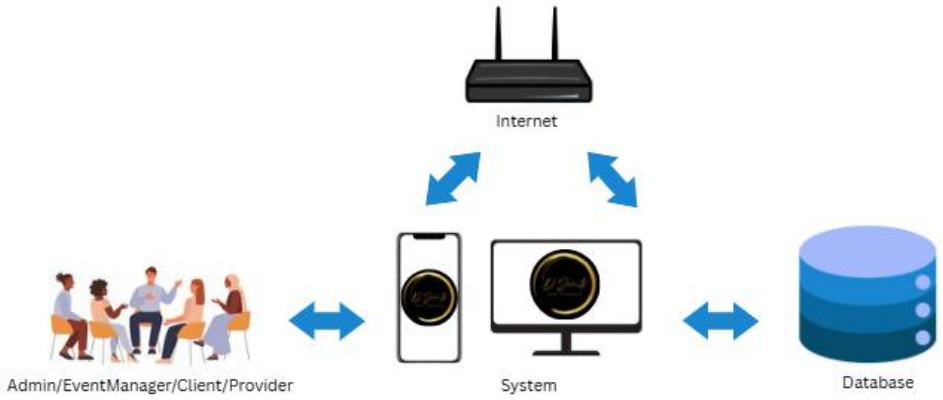
$\sum^{n}_{i=1} w_i x_i$ is the sum of the products of each value and its weight

$\sum^{n}_{i=1} w_i$ is the sum of the weights

3.8 Network Diagram

The network diagram illustrates the connectivity and data flow of the EVENTECH system. It shows how administrators, event managers, clients, and service providers access the platform using client devices such as desktop computers, laptops, and mobile phones

through the Internet. As a web-based platform, EVENTECH serves as the central system that processes user requests, manages event reservations, stores and retrieves provider information, handles service outsourcing activities, and supports communication among different users. Through this setup, the system enables efficient coordination of event management operations while maintaining centralized access to data and services.



3.8.1 Network Diagram

Figure 3.8.1 presents the network architecture of the EVENTECH system. It shows how the main users of the platform, namely the Admin, Event Manager, Client, and Provider, access the system through an internet connection using their devices. These users act as the client side of the system and interact with the web-based platform to perform their assigned functions. Through this setup, the Admin can supervise system activities, the Event Manager can manage event operations, the Client can create and monitor event requests, and the Provider can submit bids and respond to service-related opportunities.

At the center of the architecture is the EVENTECH system, which serves as the main application platform. It processes all incoming user requests and executes the necessary system functions. These functions include user authentication, role-based access control, event posting, bid management, reservation handling, payment processing,

notification delivery, and other core operations. As the central processing component, the system ensures that user actions are validated and properly managed before the corresponding data is stored or retrieved.

The database is positioned on the right side of the diagram and is directly connected to the system. It serves as the centralized storage for all system-related data, including user accounts, event posts, bids, reservations, payments, provider records, reviews, and notifications. The database supports accurate record-keeping, reliable data retrieval, and consistency of information across all modules of the platform. This centralized design helps maintain the integrity and security of the system's data.

Overall, the network architecture demonstrates how the users connect to the EVENTECH platform through the internet, how the system processes their requests, and how the database stores the necessary information. This setup supports efficient communication, centralized management, secure storage, and smooth coordination among the different user roles, contributing to a more organized and reliable event management process.